

MUHAMMAD TAYYAB

Software Engineering Student • Full-Stack Developer • AI / NLP & RAG Systems
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PROFESSIONAL SUMMARY

High-performing Software Engineering undergraduate at FAST-NUCES, Dean's List honoree, and Teaching Assistant for Operating Systems. Delivered a production-deployed full-stack platform (FBR e-Invoicing) during studies and improved a peer-reviewed ML paper's AUC from 69–70 % to 78–82 % using advanced ensemble methods. Actively building AI-powered RAG applications and semantic retrieval systems. Seeking Software Engineering, Full-Stack, or AI Engineering internships.

EDUCATION

BS Software Engineering — FAST-NUCES, Chiniot-Faisalabad

2023 – Present (Expected 2027)

Dean's List Honoree — 6th Semester | Teaching Assistant, Operating Systems (2025 – Present)

TECHNICAL SKILLS

Languages: Python • C / C++ • Java • JavaScript • SQL • Kotlin • Assembly • Shell Script

AI / NLP: PyTorch • Scikit-learn • XGBoost • LangChain • RAG Pipelines • Vector Embeddings • SHAP • OpenCV

Full-Stack: React • Next.js • Vite • Tailwind CSS • Node.js • Express • FastAPI • Redux Toolkit • REST APIs

Databases: MySQL • MongoDB • Firebase • SQLite

DevOps: Git • GitHub Actions • Docker • Vercel • Linux • JIRA • Figma • Prompt Engineering

WORK EXPERIENCE

Teaching Assistant — Operating Systems

2025 – Present

FAST-NUCES, Chiniot-Faisalabad Campus

- Facilitated weekly lab sessions for 40+ students covering process scheduling (FCFS, Round Robin, Priority), IPC, semaphores, and multi-threading; produced C/C++ reference implementations used as teaching aids.
- Designed and graded lab assignments on core OS concepts; held regular office hours for one-on-one mentorship, resulting in measurable improvements in student performance.

Front-End Developer (Project) — FBR e-Invoicing System

2025

Client Engagement — Production Deployment

- Led complete front-end delivery of a live invoicing platform: dashboards, invoice workflows, RBAC, and protected routes across multiple user tiers; built scalable React component library with Redux Toolkit + Axios.
- Delivered production-ready code on schedule during undergraduate studies — a verifiable industry milestone; system remains in active use by the client.

FEATURED PROJECTS

FAST University Newcomer RAG Assistant

2025 – Present

Stack: LangChain, Vector Embeddings, FastAPI, React, Firebase, Python

- Built a conversational RAG chatbot for FAST University newcomers — answers admissions, campus life, and course queries by retrieving from a curated FAST knowledge base, eliminating the information barrier for incoming students.

NHANES Depression Prediction — Research Replication & Improvement

2025 – Present

Stack: Python, Scikit-learn, XGBoost, Bayesian Optimization, SHAP, Pandas

- Reproduced and outperformed a **Frontiers in Computer Science (2025)** paper; improved AUC 69–70 % → 78–82 % via Ensemble Stacking + Bayesian Optimization; applied SHAP to identify top clinical predictors.

FAST_RAGger — High-Performance RAG Chatbot

2025 – Present

Stack: React, Vite, Tailwind CSS, Firebase, LangChain, Vector Embeddings

- Full-stack RAG app with document upload, sub-second semantic vector search, AI responses, Firebase auth, and a production-ready modular React/Tailwind UI — starred by the community on GitHub.

NLP & Semantic Retrieval Systems

2025 – Present

Stack: PyTorch, TF-IDF, Word2Vec, GloVe, Streamlit, NLTK

- Emotion classification (PyTorch) + hybrid semantic search (TF-IDF / Word2Vec / GloVe) with interactive Streamlit dashboards and full evaluation metrics (F1, Precision, Recall, confusion matrices).

ACHIEVEMENTS

- Dean's List — 6th Semester:** Awarded for outstanding academic performance at FAST-NUCES — top percentile of the class.
- Teaching Assistantship (OS):** Competitively selected as TA; conducted labs and mentored 40+ students on scheduling, concurrency, and systems programming.
- Research Improvement:** Outperformed published Frontiers authors on the same NHANES dataset (AUC +8–12 %), demonstrating independent research capability at undergraduate level.
- Production Deployment:** Shipped a live client-facing platform (FBR e-Invoicing) during undergraduate studies — verifiable production experience most students do not have.